20250108 AK2030 AK2036 Tentamen

Vetenskapsteori och vetenskaplig metodik (naturvetenskap) (Kungliga Tekniska Högskolan)



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Quick information

Time limit: 4 hours (6 for FUNKA students with extended time)
Allowed resources: Any books, printed and hand-written material on paper. No electronic devices, such as computers or phones, are allowed.

Final exam
Theory and Methodology of Science

Course codes: AK2030, AK2036

2025-01-08

General information

The exam consists of three parts corresponding to the learning outcomes of the course. It is an "open book" exam, which means that you are allowed to bring printed copies of the course literature or any other literature as well as your own notes. A dictionary (published paper dictionary, such as Swedish-English) is also allowed. All such material must be printed or hand-written on paper. No electronic devices, such as computers or phones, are allowed during the exam. You are allowed to cite sources, i.e. the course literature that you bring. Such citations should be put in quotation marks (" ") and a reference (using any standard system) should be given immediately after the quote. These citations will not be included in the assessment, and the content will not increase your grade. A possible use for a citation might be to exemplify a position you disagree with in your treatment of part 3.

Please write clearly when you fill out the blue front page. Otherwise, we might not be able to grade your exam. The personal number should correspond to the one on the invigilator's list. You do not need to fill in the section "Mark attempted questions with X" at the bottom of the front page.

Grading: Part 1 is a multiple-choice part; 15 questions, one point each. Several options may be correct. Points per question will be either full point or zero points. Bonus points will be added to your score on part 1, up to the maximum limit of 15 points on part 1. Bonus points from period 2, 2024 are valid for this exam.

Part 2 consists of two problems, five points each. Points will be awarded in 0.5 increments. Each problem contains several sub-problems and is to be answered in essay form. In this part, you are tasked with demonstrating knowledge by describing and accounting for the course concepts in your own words, applying these to new cases and by critically discussing these concepts. By completing only part 1 and part 2, a student can get the following grades: C, D, E, FX and F. The grade on these parts is determined by the grading table for parts 1 and 2 below.

Part 3 is about depth. Here, you get a number of questions on different topics to choose between and answer with one longer essay answer. You will argue for your opinion and justify your positions, explain and discuss the course concepts, and how

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they relate to your field of study. Part 3 will be graded A, B or C according to the grading criteria (see below). This part will only be corrected if you have received the grade C on parts 1 and 2. You are not required to answer this part to pass the exam.

Final grade. For students who complete only parts 1 and 2, the final grade is determined by the grading table for those parts. For students who receive the grade C on parts 1 and 2 and have submitted an answer to part 3, the final grade is determined by the grade on part 3.

The exam is four hours long. You are free to leave after one hour. If you have questions regarding the content of the exam, please contact the invigilator.

Examiner: Prof. Till Grüne-Yanoff, Division of Philosophy, KTH. Visiting address: Teknikringen 76.

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Grade Table Parts 1 & 2

Score parts 1+2	Grade
<14.0	F
14.0	Fx
17.0	E
19.0	D
22.0	C

Grading Criteria Part 3

Grade A. Studenten redogör korrekt, samt med stor utförlighet och precision för kursbegrepp, teorier och problemområden samt gör rimliga tillämpningar av dessa begrepp och teorier på ett mycket övertygande sätt.

Studenten framställer en välargumenterad diskussion av definitionerna och tillämpningarna av begrepp och teorier med avseende på vetenskaplig forskning på ett utförligt, självständigt och mycket precist sätt.

Translation. *The student provides correct, extensive and precise accounts for concepts, theories and general problem areas, and provides very convincing applications of those concepts and theories to specific cases.*

The student presents a well-argued, independent, extensive and very precise discussion of the definitions and applications of concepts and theories as they apply to specific cases of scientific research.

Grade B. Studenten redogör korrekt och med precision för kursbegrepp, teorier och problemområden samt gör rimliga tillämpningar av dessa begrepp och teorier på ett övertygande sätt.

Studenten framställer en huvudsakligen välargumenterad diskussion av definitionerna och tillämpningarna av begrepp och teorier med avseende på vetenskaplig forskning på ett utförligt och precist sätt samt med viss självständighet i framställningen.

Translation. *The student provides correct and precise accounts for concepts, theories and general problem areas, and provides convincing applications of those concepts and theories to specific cases.*

The student presents an extensive, precise, mostly well-argued, and somewhat independent discussion of the definitions and applications of concepts and theories as they apply to specific cases of scientific research.

Grade C. Studenten redogör korrekt och tydligt för kursbegrepp, teorier och problemområden samt gör rimliga tillämpningar av dessa begrepp och teorier på specifika fall.

Studenten framställer en diskussion av definitionerna och tillämpningar av begrepp och teorier med avseende på vetenskaplig forskning på ett precist sätt med ansats till argumentation och självständighet.

Translation. *The student accounts, correctly and clearly for concepts, theories and general problem areas, and provides reasonable applications of these concepts and theories to specific cases.*

The student presents a discussion of the definitions and applications of concepts and theories as they apply to specific cases of scientific research in a precise way with an attempt at independent and argumentative reasoning.

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PART I – Multiple choice questions

Instructions for multiple choice questions: For each question there is *at least* one correct answer alternative. In other words, for each question the number of correct alternatives is 1, 2, 3, or 4. Mark your answers with "X" on the separate answer sheet, *not* on the exam. You are awarded one point for each question if, and only if, you have selected the correct alternatives and no other. Partial points are not awarded on this part.

1. What is true about the justification of method choice? Mark all correct options.
   a) Scientists are sometimes not justified in their method choices.
   b) All ways of justifying method choice satisfy the same goals.
   c) From the perspective of the epistemic tool view of methodology, science ought to be done by choosing methods that fulfil epistemic goals.
   d) Scientists should choose the methods which give the best economic results.

2. What is true regarding hypothesis confirmation? Mark all correct options.
   a) A certain set of evidence can only support one hypothesis.
   b) If a hypothesis can be falsified deductively (if certain auxiliary assumptions are taken to be true), then its confirmation can be deductively inferred as well.
   c) Hypothesis confirmation requires fallible inductive inferences.
   d) Several hypotheses can be confirmed by the same evidence, which is problematic.

3. What is true regarding the problems with falsificationism and the hypothetico-deductive method? Mark all correct options.
   a) In scientific practice, scientists always consider all relevant non-falsified hypotheses equally likely.
   b) Given the Duhem/Quine thesis, there is no asymmetry between confirmation and falsification.
   c) The hypothetico-deductive method utilizes ad-hoc hypotheses.
   d) In scientific practice, scientists sometimes want to claim that a hypothesis is more or less likely to be true.

4. Which of the following claims about observations are true? Mark all correct options.
   a) Indirect observations require operationalizations.
   b) When you are observing the effect of an event rather than the event itself you are making an indirect observation.
   c) An aided direct observation is, for example, when you measure the political opinions of people with a survey.
   d) When your observation requires an aid, such as a telescope or microscope for observing some feature of interest with sufficient precision, it is an aided observation.

5. What is true about interval scales? Mark all correct options.
   a) Zero on the scale represents the complete absence of the property concerned.
   b) If a quantity is measured on an interval scale, it cannot also be measured on a ratio scale.
   c) Interval scales are identical up to positive linear transformations.
   d) On (linear) interval scales, arithmetic means are meaningful as averages.

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6. What is true about Randomised Control Trials (RCTs)? Mark all correct options.
   a) RCTs guarantee that background factors are equally distributed between treatment and control group.
   b) Subjects are separated into treatment and control group by a random process.
   c) Stratification can help when one wants to distribute unknown background variables between treatment and control group in RCTs.
   d) RCTs facilitate both single and double blinding.

7. What are necessary aspects of scientific experiments? Mark all correct options.
   a) An observational act of some kind (direct, aided-direct or indirect).
   b) A statistical model used to infer results.
   c) A laboratory in which the experiment takes place.
   d) An intervention.

8. What is true about isolating models? Mark all correct options.
   a) Isolation models can only be constructed theoretically, as calculations or in a computer simulation.
   b) The idea behind isolation models is that we can pick out some properties of the target and model these in isolation from other properties.
   c) An isolation model is less transparent than a mirror model of the same target.
   d) An isolation model represents a subset of the target's workings and/or structure.

9. What is true about the relation between models and experiments? Mark all correct options.
   a) External validity is more of a problem for experiments than for models.
   b) Models always contain purposeful idealizations, experiments do not necessarily include such idealizations.
   c) Internal validity is generally less of a problem for models than for experiments.
   d) One similarity between experiments and models is that they always are mere representations of targets.

10. Which claims are true about Bayesian statistics? Mark all correct options.
    a) Bayesian statistics can evaluate a large set of competing hypotheses.
    b) The "problem of old evidence" remains a substantial problem for Bayesian statistics.
    c) Bayesian statistics offers an unambiguous algorithm for assigning prior probabilities to hypotheses.
    d) Bayesian statistics is not viable way of modelling human belief change, as compared to the statistical frameworks of Fisher and Neyman-Pearson.

11. What is true regarding the structure of explanations? Mark all correct options.
    a) Successful explanations are contrastive, meaning that they explain why X happened while simultaneously explaining why Y did not happened.
    b) An explanans can have a particular form and thereby explain a particular explanandum, without being able to explain similar, but different, phenomena.
    c) An explanans can have a general form (i.e. law-like regularities) from which a particular explanandum is derived.
    d) An explanation features an explanandum, something to be explained, and an explanans, something which provides the explanation.

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12. What is true of tacit knowledge? Mark all correct options.

    a) The ability to ride a bicycle is an example of tacit knowledge.
    b) Tacit knowledge can sometimes be exemplified as an ability to communicate a capacity or skill by demonstration - i.e. without first having to study the capacity or skill.
    c) Tacit knowledge is taught mainly through engineering textbooks and manuals.
    d) Being able to fluently apply the quadratic formula to solve quadratic equations is an example of tacit knowledge.

13. What is true about belief-desire explanations? Mark all correct options. Belief-desire explanations ...

    a) ... are impossible to improve due to their subjective nature.
    b) ... are never viewed as causal explanations.
    c) ... are used to explain human behavior.
    d) ... refer to mental states in their explanantia.

14. A decision has to be made between three alternative courses of action.
    On alternative A, there are two possible outcomes: 100 people dying (probability 0.6) or no one dying (probability = 0.4).
    On alternative B, there are four possible outcomes (equal probabilities): 20 people dying, 100 people dying, 400 people dying, or 4 people dying.
    On C, there are two possible outcomes: 800 people dying (probability 0.01) or no one dying (probability 0.99).
    Mark all correct options, given the expected value criterion for decisions involving risks:

    a) C is a better alternative than A.
    b) A is the second best alternative.
    c) B is the third best alternative.
    d) C is the second best alternative.

15. What is true about the *precautionary principle*? Mark all correct options. The precautionary principle ...

    a) ... states that experiments exceeding the research budget shouldn't be conducted.
    b) ... states that if an action or policy has a suspected risk of causing harm to the public or to the environment, in the absence of scientific consensus that the action or policy is harmful, the burden of proof that it is not harmful falls on those taking the action.
    c) ... states that the burden of proof of an activity being harmless falls on the proponent of the activity, rather than on the public.
    d) ... is of no concern in ideal democracies.

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PART II – Assignment for grades up to C

Instructions for Part II: *This part consists of two short answer questions. You should answer both questions. Your task is to demonstrate knowledge by describing the course concepts in your own words, applying these to new cases and by critically discussing these concepts. The questions in part 2 will test skills up to grade level C. You should solve the two problems on separate sheets of paper and only write on the front side of each paper. Expected length per answer is about one page – longer than a couple of sentences, but shorter than two full pages, depending on your handwriting – but there are no formal requirements regarding length.*

1. **Problem 1. (5 points)** Give two examples of measures: one that has a natural zero and one that does not. Explain what kind of scale is commonly used for each of them and why.

   Suppose that you are observing the occurrence of a certain kind of infection by a bacterial culture test – i.e. by taking a sample from the patient's body and placing it in a culture medium, in this case so-called nutrient agar. Explain what the relevant operationalization is and explain how one could test convergent validity by also growing the bacteria in a different medium, so-called plate count agar (don't worry about the chemical details of these two media – the only relevant point here is that they are different).

   Random and systematic error is discussed in this course as an issue that arises when measuring quantities. Discuss whether something similar to these errors can be applied to qualitative observations as well and why you think that we commonly relate them to quantities rather than qualities.

2. **Problem 2. (5 points)** Provide a possible DN explanation for the explanandum "The water boiled at a 100 degrees C down in the valley but not at the top of the mountain". (What's important here is the structure, not complete accuracy of the explanans.)

   Explain the problem of symmetry concerning DN explanations and motivate why your explanation is or isn't subject to that problem.

   Consider the following explanation. (E): "Our unmanned aircraft crashed because the way we programmed the stabilization system didn't consider strong winds." Discuss one way to improve the explanandum of E and one way to improve the explanans of E. Connect your improvements to some of the explanatory virtues.

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PART III – Assignment for grades A-C

Instructions for Part III: *This part is about depth. Here, you choose one of the topics below and answer with a longer essay answer. You will be graded not only on the extent to which you demonstrate proficiency with the course content, but also on the quality and coherency of the justification you provide for taking a certain position, etc. This assignment will only be corrected if you have been graded C on parts 1 and 2. Grading criteria can be found in the general information. You should only write on the front side of each paper. Expected length is about 500-1000 words, but there are no formal requirements regarding length.*

Choose one of the following problems to answer.

1. **Problem 1.** The second step of the hypothetico-deductive (HD) method is to deduce observable consequences from the hypothesis. What problems might arise in this step? Are these deductions always straightforward or might one draw the wrong conclusions? Come up with at least one problematic example.

   Next, compare this to the problem, as outlined by the Duhem-Quine (DQ) thesis, when making a Modus Tollens inference in step four of the HD method. Discuss how these problems might affect the conclusions. Is one problem worse than the other?

2. **Problem 2.** Describe what an experiment is according to the course literature, by giving an example and pointing out the key feature(s) as well as typical characteristics of such a scientific activity (you may construct an imaginary example or use a real scientific case, but not one from the course literature).

   Next, carefully explain the control methods control group randomization and blinding, , and how they could be used in your example from above. Discuss why the methods should be used; what control problems could be dealt with/handled/addressed by them, and to what degree.

   Next, explain what a natural experiment and a simulation is, and give an example of both.

   Finally, discuss if natural experiments and simulations should be regarded as proper experiments or not, given your earlier characteristics of what an experiment is. Do you agree with these characteristics? Could there be any problems with these characteristics related to simulations or natural experiments? Make sure you carefully explain the concepts you discuss and justify your reasoning.

3. **Problem 3.** Alex is part of a research group and discovers that another member, Billie, has committed research fraud. While considering what to do, Alex comes up with the following.

   > If I tell the media about the fraud, many people will read about it and will get a more negative view on science in general. If I tell the research director instead, this will be handled internally and probably never reach the media. In both cases, Billie will most likely get fired. Also, committing fraud is a form of lying – and one shouldn't lie. But I am not sure whether it counts as lying if I do not reporting Billie; after all, I wasn't the one committing the fraud. At the same time, doing

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> nothing about the fraud feels wrong and I think that other people in a similar situation would take their responsibility and act.

It is not clear which ethical framework Alex's reasoning fits with. Discuss which parts of Alex's reasoning seem to appeal to deontology, virtue ethics and consequentialism respectively. Further, discuss what would be the ethical thing for Alex to do in the situation from the perspective of at least two of the ethical frameworks. Part of your discussion should be a critical evaluation of Alex's own reasoning.

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